THE UK UNIVERSITY INTEGRATION BEE 2024/2025

$$\int$$

ROUND TWO
CROSSNUMBER ROUND (DOWN)

Sponsored by



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DOWN

1. Three distinct digits a, b, c arranged in some order, where b and c are both also digits of a^3 and $|a-b|^3$.

3. $\left|20\displaystyle\int_1^3 f(x)\,\mathrm{d}x - 25\int_0^1 f(x)\,\mathrm{d}x\right|$, given that

$$20\int_0^1 f(x)\,\mathrm{d}x + 25\int_0^2 f(x)\,\mathrm{d}x = 2025$$

and

$$25\int_1^2 f(x)\,\mathrm{d}x + 20\int_0^3 f(x)\,\mathrm{d}x = 2520.$$

5. The value of a such that

$$\int_2^3 \frac{ax+5}{x^2-6x+10}\,\mathrm{d}x = 20\pi - \frac{25\ln(2)}{2}.$$

6. A palindromic number which is divisible by 11.

9. Middle digit $= \left\lfloor \dfrac{\text{Largest answer}}{\text{Second largest answer}} \right\rfloor$.

10. The only integer value in the set $\left\{ \left(\dfrac{1}{\pi^k}\displaystyle\int_0^\pi \left(\sum_{n=1}^{\infty} \frac{\sin(nx)}{n} \right)^{\frac{(2\text{ ACROSS})}{(1\text{ ACROSS})}-5} \mathrm{d}x \right)^{-3} : k \in \mathbb{Z} \right\}$.

12. $\displaystyle\int_0^{(7\text{ ACROSS})+(2\text{ ACROSS})} \lfloor x\lfloor x\rfloor\rfloor - \left\lfloor \sqrt{x\lfloor x\rfloor} \right\rfloor^2 \mathrm{d}x - 2050.$

13. A 4-digit number $\overline{abcd}$ where

    - d is half the value of the second-smallest digit (and is not 0), and
    - $\displaystyle\lim_{n\to\infty}\left(\int_{-\infty}^{\infty}\left(e^{-x^2+x}\right)^n \mathrm{d}x\right)^{\frac{1}{n}} = \frac{b-a}{2}\sqrt[\sqrt{a+b}]{e}$.

14. Of 3 DOWN and this number, one number divides the other. They might even divide each other. Who knows?

In the answers to the following clues, the digits 1-9 appear once and exactly once.

15. Not a prime number.

16. Last digit is 3 less than the first digit of 15 DOWN.

17. An odd 3-digit number $\overline{abc}$ with $a < b < c$ and $a + b + c = 20$.